

Math 208 - Tutorial 3

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1 IT Center Scheduling (Question 2.15)

We are looking to schedule staff for an IT Center. The center is open from 8 am until midnight. We are given the following table on staffing requirements:

Time Interval	Staff Required
8:00 - 12:00	4
12:00 - 16:00	8
16:00 - 20:00	10
20:00 - 00:00	6

There are two types of consultants: full-time and part-time.

- Full-time staff are paid \$14/h and can work any of the three shifts:
 - 8:00 - 16:00 (morning)
 - 12:00 - 20:00 (afternoon)
 - 16:00 - 00:00 (evening)
- Part-time consultants are paid \$12/h and can work any of the 4 shifts listed in the table above.

During every time period, there must be as many full-time consultants or more as there are part-time consultants on duty.

Determine the minimum cost staffing plan for the center.

Solution

Problem Parameters

The problem parameters are the number of staff required for each shift and the cost of full-time and part-time employees. Let m_i be the staff required for the time interval i , $i \in \{1, 2, 3, 4\}$ with each element in the set respectively corresponding to a Time Interval in the table. Let c_f and c_p be the hourly cost for a full-time and part-time employee respectively.

Problem Decision Variables

The decision variables are how many full-time and part-time staff we have for each shift. Let f_i be the number of full-time employees in shift i , $i \in \{1, 2, 3\}$ representing the morning, afternoon, and evening shifts. Let p_i be the number of part-time employees in the time interval i , $i \in \{1, 2, 3, 4\}$, with i representing the time of day as shown in the table above.

Problem Constraints

The first constraint is that we must have the required amount of staff in each time interval.

$$\begin{aligned}f_1 + p_1 &\geq m_1 \\f_1 + f_2 + p_2 &\geq m_2 \\f_2 + f_3 + p_3 &\geq m_3 \\f_3 + p_4 &\geq m_4\end{aligned}$$

The second constraint is that in each time interval, there must be as many full-time consultants as part-time consultants.

$$\begin{aligned}f_1 &\geq p_1 \\f_1 + f_2 &\geq p_2 \\f_2 + f_3 &\geq p_3 \\f_3 &\geq p_4\end{aligned}$$

Problem Objective

The problem objective is to minimize the total cost of the schedule.

$$\min \quad 8c_f \sum_{i=1}^3 f_i + 4c_p \sum_{i=1}^4 p_i$$

Full Model

The full model is as follows:

$$\begin{aligned} \min \quad & 8c_f \sum_{i=1}^3 f_i + 4c_p \sum_{i=1}^4 p_i \\ \text{Subject To:} \quad & f_1 + p_1 \geq m_1 \\ & f_1 + f_2 + p_2 \geq m_2 \\ & f_2 + f_3 + p_3 \geq m_3 \\ & f_3 + p_4 \geq m_4 \\ & f_1 \geq p_1 \\ & f_1 + f_2 \geq p_2 \\ & f_2 + f_3 \geq p_3 \\ & f_3 \geq p_4 \end{aligned}$$

The optimal solution has a minimized scheduling cost of \$1456. The shifts are as follows:

Shift	Staff
f_1	2
f_2	2
f_3	3
p_1	2
p_2	4
p_3	5
p_4	3

There are no shifts that are overstaffed, and there are as many full-time consultants as there are part-time consultants for each shift. All the constraints hold with equality. The hours 12:00 - 16:00 and 16:00 - 20:00 require a lot more part-time staff than the other time slots.

2 Scheduling TV Production (Question 2.13)

An electronics company manufactures three types of TVs: plasma, LED, and 3D. Each type of TV requires different production times as shown in the table below:

	Plasma (PL)	LED	3D	Capacity
Components	36	50	60	15,000
Assembly	20	36	50	10,500
Test	30	15	40	12,000

The profits for each product are \$90 for plasma, \$120 for LED, and \$175 for 3D. Due to existing commitments, at least 10% of TVs manufactured have to be plasma, 20% LED, and 15% 3D.

Solution

Let $I = \{\text{PL}, \text{LED}, \text{3D}\}$ be the set of TV types.

Problem Parameters

The problem parameters are the time it takes for each component, the capacity in each department, the proportion of each TV that has to be produced, and the cost of each TV. Let c_i , a_i , t_i , $i \in I$ be the time required in components, assembly, and test for each TV brand. Let C , A , T be the capacity of each department. Let q_i , $i \in I$ be the proportion of TVs of type i that has to be produced. Let p_i , $i \in I$ be the profit for each TV.

Problem Decision Variables

The decision variables are how many TVs of each type to produce. Let x_i , $i \in I$ be the number of units of each TV type to produce.

Problem Constraints

The first set of constraints is the time capacity within each department:

$$\begin{aligned}\sum_{i \in I} c_i x_i &\leq C \\ \sum_{i \in I} a_i x_i &\leq A \\ \sum_{i \in I} t_i x_i &\leq T\end{aligned}$$

The second set of constraints is the proportion of TVs that must be produced:

$$\frac{x_j}{\sum_{i \in I} x_i} \geq q_j, \quad j \in I$$

This is a blending model, which can be rearranged to a linear form:

$$x_j - q_j \left(\sum_{i \in I} x_i \right) \geq 0, \quad j \in I$$

Problem Objective

The problem objective is to maximize the profit from the TVs sold:

$$\max \sum_{i \in I} p_i x_i$$

Full Model

The full model is as follows:

$$\begin{aligned} & \max \quad \sum_{i \in I} p_i x_i \\ \text{Subject To:} \quad & \sum_{i \in I} c_i x_i \leq C \\ & \sum_{i \in I} a_i x_i \leq A \\ & \sum_{i \in I} t_i x_i \leq T \\ & x_j - q_j \left(\sum_{i \in I} x_i \right) \geq 0, \quad j \in I \end{aligned}$$

The maximum profit for the TVs is \$39,619.56. The production levels are as follows:

TV Type	Production Level
PL	163.04
LED	65.21739
3D	97.82

In total, 50% of the TVs produced were plasma, 20% LED, and 30% 3D. The proportion of LED TVs is at equality for its constraint, suggesting that LED TVs are not as profitable as plasma or 3D TVs. The constraints are at capacity for components and assembly, while we have more than 2000 testing hours left available.